CIA-1

MLA0205

Fundamentals of Machine Learning

i) $P(+) = 0.03$, Sensitivity $P(+|c) = 0.90$ Specificity $P(-|\neg c) = 0.94$

$P(+|\neg c) = 1 - 0.94 = 0.06$

$P(+|c) = 0.10$ : $P(-|c) = 0.90$

ii) $P(+) = 0.90 \times 0.03 + 0.06 \times 0.97 = 0.027 + 0.0582 = 0.0852$

$P(-) = 1 - 0.0852 = 0.9148$

iii) $P(c|+) = 0.0270/0.0852 = 0.3169$; $P(\neg c|+) = 0.6831$; $P(c|-) =$

$(0.10 \times 0.03)/0.9148 = 0.0033$

$P(\neg c|-) = 0.9967$

iv) $R(\text{treat}|+) = 2 \times 0.6831 = 1.3662$; $R(\text{no treat}|+) = 50 \times 0.3169 = 15.845$

$R(\text{treat}|-) = 2 \times 0.9967 = 1.9934$; $R(\text{no treat}|-) = 50 \times 0.0033 = 0.1650$

v) Positive → treat ($1.3662 < 15.845$); negative → do not treat ($0.1650 < 1.9934$)

$P(c|x) > 2/(2+50) = 0.0385$. $P(c|+) = 0.3169$ $P(c|-) = 0.0033$ does not

vi) $R = 0.0852 \times 1.3662 + 0.9148 \times 0.1650 = 0.1164 + 0.1500 = 0.2664$

vii) posterior odds $= (0.3169/0.6831) \times (0.90/0.06) = 0.4639 \times 15 = 6.959$;

$P(c|++) = 6.959/7.959 = 0.8744$

viii) decide c only if $P(c|x) > 0.5$ since $P(c|+) = 0.3169 < 0.5$

the rule never treats.

the minimum-risk rule reduces cost from 1.5 to

0.2664 / a reduction of 82.24%.

2)

i) $P(x) = (0.32, 0.38, 0.30)$; $P(y) = (0.68, 0.32)$

ii) $H(x) = -\sum p \log_2 p = 0.5260 + 0.5304 + 0.5211 = 1.5776$ bits,

$P(y) = 0.3784 + 0.5260 = 0.9044$ bits

each termin $-p \log_2 p$;

iii) $H(x,y) = 0.5211 + 0.1129 + 0.5142 + 0.3322 + 0.3322 + 0.6644$

$= 2.2770$ bits

iv) $H(y/x) = H(x,y) - H(x) = 2.2770 - 1.5776 = 0.6994$ bits

v) $I(x,y) = H(y) - H(y/x) = 0.9044 - 0.6994 = 0.2050$ bits

vi) $H(P_x, Q) = -\sum P_x \log_2 Q = 0.32(\log 1.0000) + 0.38(1.7370) + 0.30(2.3219)$

$= 1.6766$ bits

vii) KL divergence $D_KL(P_x \| Q) = H(P_x, Q) - H(x) = 1.6766 - 1.5776$

$= 0.0990$

viii) $P(y = yes \mid x = high) = 0.20/0.30 = 0.6667$ (low: 0.0625; medium: 0.2632

$P(yes) = 0.32$

iii) $S^2 = 46/5 = 9.20$ ; $S = 3.033$

dividing by $n-1$ corrects the bias ($E[\sigma^2 - \mu^2] = ((n-1)/n)\sigma^2 = (5/6)\sigma^2$)

iv) $P(x > 59) \cdot 2 = (59 - 51) / 3.69 = 1.445$ ; $P = 1 - \Phi(1.445) = 0.0743$

table values $\Phi(1.44) = 0.9251$ and $\Phi(1.45) = 0.9265$ gives $0.074 - 0.075$

with $S = 3.033$ the tail becomes $0.0936$

v) $P(48 < x \leq 54) \cdot 2 \pm (31.49) = \pm 1.683$ $P = 2\Phi(1.083) - 1 = 8.7214 \approx 6.72$

Accept $6.72 \pm 0.01$ using $\Phi(1.08) = 0.8599$

b)

i) $\bar{x} = 30$; $\bar{y} = (45 + 38 + 52 + 61 + 79)/5 = 51$ $S_{xx} = \Sigma (x - \bar{x})^2 =$

$400 + 100 + 0 + 100 + 400 = 1000$

$S_{xy} = \Sigma (x - \bar{x})(y - \bar{y}) = 520 + 130 + 0 + 100 + 560 = 1310$

ii) $w_1 = 1310/1000 = 1.31$ MS per $\%$. $w_0 = 51 - 1.31 \times 30 = 11.7$ hs

$\hat{y} = 11.7 + 1.312$

iii) $24.8, 37.9, 51.0, 64.1, 77.2$ residual ; $0.2, 0.1, 1.0, -3.1, 1.8$ ; $SSE = 0.04 + 0.01$

$+1.00 + 9.61 + 3.24 = 13.90$

iv) $SST = 676 + 169 + 1 + 100 + 784 = 1730$; $R^2 = 1 - 13.90/1730 = 0.9920$

v) $w_1 = S_{xy}/S_{xx} = 1310/1500 = 0.8733$; $w_0 = 51 - 0.8733 \times 36 = 20.8$

vi) $\hat{y}(60) = 20.8 + 52.4 = 77.2$ MS ; SSE = 20458 ; $R^2 = 0.8817$

vii) shrinks slope by 33.3% SSE increase about 14.7 times

4) i) Prior $P(spam) = 40/100 = 0.40$ ; $P(legit) = 0.60$

ii) offer = 0.6905 ; free 25/42 = 0.5952 ; meeting = 34/62 = 1190

iii) offer = 7/62 = 0.1129 ; free 4/62 = 0.0645 ; meeting 34/62 = 0.5484

iv) $x = (offer = 1, free = 1, meeting = 0)$ Spam = $0.60 \times 0.6905 \times 0.5952 \times (1 - 0.1190)$

$= 0.1448$; legit $= 0.60 \times 0.1129 \times 0.0645 \times (1 - 0.5484) = 0.00197$

v) $P(spam|x) = 0.1448 + 0.00197) = 0.9866$ ; $P(legit|x) = 0.0134$

vi) $P(spam|x) > P(legit|x)$ i.e $P(spam|x) > 5/6 = 0.8333$. Since $0.9866 > 0.8333$ classify as spam.

vii) Accuracy = $(45 + 130)/200 = 87.5\%$ Precision = $45/(45+10) = 81.82\%$

recall = $45/(45+15) = 75\%$ $F1 = 2PR(P+R) = 78.26\%$

$F1 = 2TP/(2TP + FP + FN) = 90/115 = 0.7826$

viii) Total cost = $5 \times FP + 1 \times FN = 5 \times 10 + 15 = 65$ units

False Positive rate = $10/140 = 7.14\%$

$= 0.4(0.5) + (0.2)(1.0) + 0.1 = 0.100$; $z_2 = 0.3(0.4) + 0.5(1.0) - 0.1 = 0.550$

ii) $h_1 = \sigma(0.100) = 0.5250$; $h_2 = \sigma(0.550) = 0.6341$

iii) $z_0 = 0.6(0.5250) - 0.4(0.6341) + 0.2 = 0.2813$; $y = \sigma(0.2813) = 0.5550$

iv) $E = \frac{1}{2}(1 - 0.5550)^2 = \frac{1}{2}(0.4350)^2 = 0.0946$

v) $\delta_0 = (y - t) \cdot y(1-y) = (-0.4350)(0.5550)(0.4350) = -0.1069$

$= 0.2468$

vi) $\delta_1 = \delta_0 v_1 h_1 (1 - h_1) = (0.1069)(0.6)(0.2494) = -0.0160$; $\delta_2 = \delta_0 v_2 h_2 (1 - h_2) = (-0.1069)(-0.4)(0.2320) = 0.0099$ 0.6033

vii) $v_1 = 0.6 - 0.5(-0.1069)(0.5250) = 0.6281$; $v_2 = -0.4 - 0.5(-0.1069)(0.6341) = -0.3661$; $b_3 = 0.2 + 0.5(0.2494) = 0.2523$ 0.8934) not

viii) $w_{11} = 0.4 - 0.5(-0.0160)(0.5) = 0.4040$; $w_{12} = -0.2 - 0.5(-0.0160)(1.0) = -0.1920$

$b_1 = 0.1 + 0.0080 = 0.1080$; $w_{21} = 0.3 - 0.5(0.0099)(0.5) = 0.2975$; $w_{22} = 0.5 - (0.0099)(1.0) = 0.4950$

$b_2 = 0.1 - 0.0050 = -0.1050$

ix) $h_1 = \sigma(0.1080 + 0.4040 \times 0.5 - 0.1920) = 0.5295$; $h_2 = 0.6315$; $z_0 = 0.6281(0.5295) - 0.3661(0.6315) + 0.2535 = 0.3548$; $y = 0.5878$; $E = \frac{1}{2}(0.4122)^2 = 0.0850$